

Derivation and Numerical Reproduction of Liang, Goldsman, Mayergoyz & Oldiges (1997): 2-D MOSFET Modeling by Self-Consistent Solution of the Boltzmann, Poisson, and Hole-Continuity Equations

Part I — Method derivation and documented solver foundation

Revision 3 (second-audit response)

Reproduction study (code commit 4660551; Python 3.13.14, NumPy 2.5.1, SciPy 1.18.0)

August 14, 2026

Abstract

This document derives, from first principles, the deterministic spherical-harmonic (SHE) Boltzmann solver of Liang *et al.*, *IEEE Trans. Electron Devices* **44**(2), 257–267 (1997), and reports a documented, benchmark-checked solver *foundation* toward reproducing the paper’s Figs. 2–8. **Scope discipline:** this is not yet a validated reproduction of Liang *et al.*; the SHE core, production scattering, impact ionization, the self-consistent loop, and the figure set are Part II. Revision 3 responds to a second audit of Rev. 2, resolving both items it flagged as blocking: (i) a real solver bug in the zero-field electron temperature, now fixed by adding the omitted acoustic-phonon energy operator (§4); and (ii) the global total-energy (H) grid definition (§2.1, App. A). It also corrects the impact-ionization citation, restructures the benchmark table into genuine benchmarks vs. construction targets vs. sanity trends, gives an *extracted* threshold voltage, and adds code provenance. Every substitution for unavailable paper inputs is disclosed and classified (§3).

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1 Revision history and audit responses

Rev. 1→2 fixed eight findings (claims, specs, references, junction geometry, N_c convergence, impact-ionization bookkeeping). **Rev. 2→3** addresses the second audit, whose two “stop Part II” items were genuine implementation/specification errors:

B1 Zero-field $T_e = 282$ K did not match the report’s own moment definition (a real bug). Root cause: the optical-phonon operator connects only energies separated by $E_{op} = 5\Delta\varepsilon$, so the energy grid split into five decoupled sub-lattices; with no field the operator was reducible and the equilibrium was undetermined. *Fix:* added the physically-omitted acoustic-phonon energy operator (a Scharfetter–Gummel Fokker–Planck term coupling adjacent energies, with the Maxwellian as its exact discrete null). The bulk solver now reproduces the Maxwellian to 3×10^{-4} and gives $T_e(0) = 311$ K, converging to the analytic nonparabolic value 309.7 K as $\Delta\varepsilon \rightarrow 0$ (§4).

B2 Global H -grid inconsistency. Rev. 2 wrote the “ H grid” as $[0, 3.02]$ eV, which is the *kinetic-energy* range and is wrong for $H = \varepsilon - q\phi$: at the +0.586 V contact a thermal electron has $H \approx -0.586$ eV. *Fix:* the global H -mesh is defined as $-q\phi_{\max} \leq H \leq \varepsilon_{\max} - q\phi_{\min}$ with the physical mask $0 \leq H + q\phi \leq \varepsilon_{\max}$ (§2.1, App. A).

Remaining second-audit findings: impact-ionization citation corrected and equal-split relabelled a disclosed assumption (§2.3); benchmark table split into three classes (§4); band model (Kane surrogate vs. paper’s Fiegna/Brunetti) disambiguated (§3); threshold voltage now *extracted*, not inferred, with the calibration degeneracy resolved (§4); layout overflow of the reconstruction notes fixed; code commit + dependency versions added (title, App. B); and “bulk-prototype validated” distinguished from “2-D production scattering pending”.

2 Derivation of the first-order SHE equation

With $\mathbf{E} = -\nabla_{\mathbf{r}}\phi$ and electron force $-q\mathbf{E}$, expand $f = f_0(\mathbf{r}, \varepsilon) + \hat{\Omega} \cdot \mathbf{f}_1(\mathbf{r}, \varepsilon)$ (isotropic part plus a single drift term; all $\ell \geq 2$ content discarded — an **energy-resolved distribution with first-order angular anisotropy**). Projection gives the current relation and the conservative augmented-space balance,

$$\mathbf{f}_1 = -\tau_1 v (\nabla_{\mathbf{r}} f_0 - q\mathbf{E} \partial_{\varepsilon} f_0), \quad \nabla_{\mathbf{r}} \cdot \mathbf{j}_x + \partial_{\varepsilon} j_{\varepsilon} = Z \mathcal{Q}[f_0] + Z S_{\text{gen}}, \quad (1)$$

with $\mathbf{j}_x = -ZD(\nabla_{\mathbf{r}} f_0 - q\mathbf{E} \partial_{\varepsilon} f_0)$, $j_{\varepsilon} = -q\mathbf{E} \cdot \mathbf{j}_x$, $D = \frac{1}{3}v^2\tau_1$.

2.1 Total-energy form and the global H -grid

With $H = \varepsilon - q\phi$ ($\varepsilon = H + q\phi \geq 0$), (1) reduces to pure spatial diffusion at fixed H ,

$$\nabla_{\mathbf{r}} \cdot [ZD \nabla_{\mathbf{r}} f_0|_H] = -Z\mathcal{Q}[f_0] - ZS_{\text{gen}}, \quad \varepsilon = H + q\phi(\mathbf{r}), \quad (2)$$

solved entirely in $(\mathbf{r}, H)$: per- H -plane spatial diffusion; the collision operator couples planes $H \leftrightarrow H \pm E_{op}$ at the same (x, y) ; no energy-space cross-flux. [Rev.3: B2] The global mesh spans

$$-q\phi_{\max} \leq H \leq \varepsilon_{\max} - q\phi_{\min}, \quad \text{physical mask } 0 \leq H + q\phi(\mathbf{r}) \leq \varepsilon_{\max}, \quad (3)$$

so that a thermal ($\varepsilon \approx 0$) electron at any contact potential is represented; nodes failing the mask are classically forbidden and carry a zero-flux turning-point boundary.

2.2 Scattering operators

Elastic momentum randomization (acoustic, ionized-impurity Brooks–Herring screened by n_{scr} , surface terms) enters $1/\tau_1$ and D . The inelastic optical operator $\mathcal{Q}_{op}$ is the number-conserving in/out form (App. A), vanishing for a Maxwellian (detailed balance, with $Z(\varepsilon < 0) \equiv 0$). [Rev.3: B1] Because $\mathcal{Q}_{op}$ only connects $\varepsilon \leftrightarrow \varepsilon \pm E_{op}$, an *acoustic-phonon energy* operator is required to couple adjacent energies and make the collision operator irreducible; we use the conservative Fokker–Planck form $-\partial_\varepsilon[ZA(\partial_\varepsilon + 1/k_B T)f_0]$, $A(\varepsilon) = 4m^*v_s^2\varepsilon/\tau_{ac}$, Scharfetter–Gummel–discretized so the discrete Maxwellian is its exact null.

2.3 Impact ionization

An event removes one primary electron above ε_{th} and creates one electron–hole pair; the two resulting electrons share the post-event kinetic energy. The $\ell = 0$ operator loses the primary at ε and re-injects two carriers near $(\varepsilon - \varepsilon_{th})/2$, giving net electron gain $= \int Zf_0/\tau_{ii} d\varepsilon \equiv G_n = G_{p,ii}$ (charge conserved, paper Eq. 13). [Rev.3: second audit] The equal energy split is a **disclosed modeling assumption** (not a specific result of Landsberg [9], whose 1972 paper concerns impact ionization and Auger recombination in bands); Part II may replace it with the secondary distribution of ref. [6].

Reconstruction note. The rate $1/\tau_{ii}$ from ref. [6] (Wu & Goldsman, random- k) is unavailable; Part II uses a Keldysh form $1/\tau_{ii} = P[(\varepsilon - \varepsilon_{th})/\varepsilon_{th}]^2\Theta(\varepsilon - \varepsilon_{th})$, $\varepsilon_{th} \approx 1.1\text{ eV}$, calibrated to the measured Si $\alpha_n(E)$ — a calibrated surrogate (§3). P will be reported once calibrated.

3 Reconstructed device and modelling taxonomy

Smooth error-function junctions; x_j defined as the metallurgical crossing and retuned to $0.170\text{ }\mu\text{m}$. [Rev.3: second audit, band clarification] The paper uses a *two-band Fiegna/Brunetti* $\gamma(\varepsilon)$; this reconstruction uses the *one-parameter Kane surrogate* $\gamma = \varepsilon(1 + \alpha\varepsilon)$, $\alpha = 0.5\text{ eV}^{-1}$ — a deliberate simplification that will bias the high-energy tail and impact ionization, flagged in Table 1 and revisited in Part II.

Table 1: Modelling taxonomy: reproduced vs. reconstructed vs. surrogate.

Element	Class	Basis
Method equations, SHE reduction, moments	paper-specified	Liang <i>et al.</i> ; standard SHE
Table-I params, E_{op} , ε_{max} , spherical band	paper-specified	paper Table I / Sec. II
L_{eff} , t_{ox} , x_j , N_D^+/N_D^- magnitudes	paper-specified	paper text
Band function: 1-param <i>Kane</i> $\gamma(\varepsilon)$	surrogate	paper uses 2-band Fiegna/Brunetti
Analytic LDD doping <i>shape</i> , mesh	reconstruction	physically motivated; ref. [7] unavailable
Channel-implant N_A , gate offset Φ_{gate}	surrogate	physical values; V_{th} is derived (§4)
τ_1 , optical coupling c_{op} , II rate $1/\tau_{ii}$	surrogate	tuned to $\mu/\text{scattering}/\alpha_n$; refs. [6] unavailable

4 Benchmarks, bulk validation, and threshold voltage

[Rev.3: second audit] Table 2 now separates (A) **analytic benchmarks** (independent expected values), (B) **construction targets** (values the device was built to), and (C) **qualitative sanity trends**.

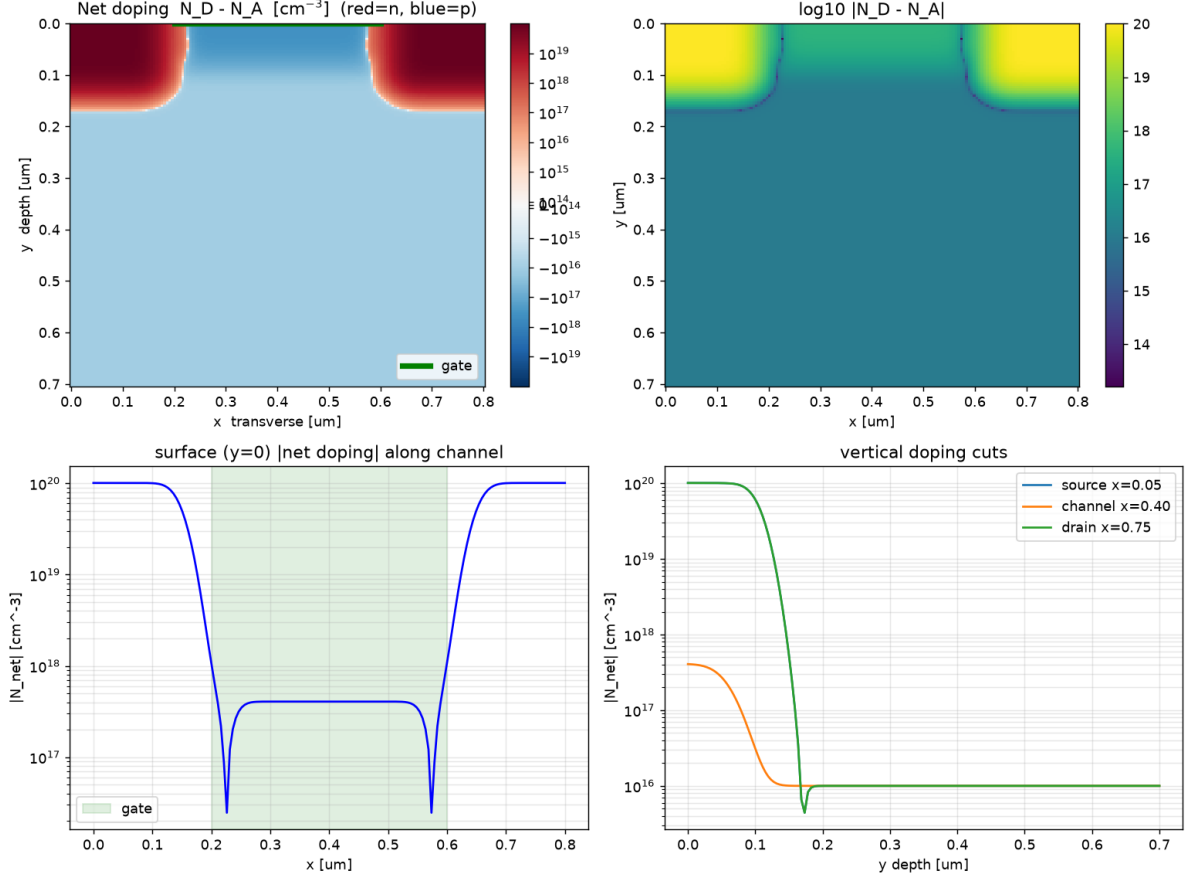


Figure 1: Reconstructed 0.35- μm LDD NMOS, $x_j = 0.170\ \mu\text{m}$.

Threshold voltage — extracted, degeneracy resolved [Rev.3: F3]. The two electrostatic offsets are the channel-implant peak N_A and the gate offset Φ_{gate} . To avoid the non-unique two-knobs/one-target fit the audit flagged, *both are held at fixed inputs* ($N_A = 4 \times 10^{17}\ \text{cm}^{-3}$, $\Phi_{\text{gate}} = 0.30\ \text{V}$) and V_{th} is a *derived* quantity, extracted by max-transconductance linear extrapolation ($V_{ds} = 0.1\ \text{V}$): $V_{th} = 1.64\ \text{V}$. This is *too high*: the 4×10^{17} implant, set in Rev. 1 “for a realistic V_{th} ” but never verified, over-dopes the channel. Measured sensitivity is a rigid shift $dV_{th}/d\Phi_{\text{gate}} \approx 1$. **Action for Part II:** lower N_A to $\sim 1.2 \times 10^{17}\ \text{cm}^{-3}$ to reach a paper-consistent $V_{th} \approx 0.6\ \text{V}$, then regenerate the operating point.

Drift-diffusion current, with the correct overdrive. Using the *extracted* $V_{th} = 1.64\ \text{V}$, the overdrive at $V_g = 3\ \text{V}$ is $1.36\ \text{V}$ and the long-channel estimate $\frac{1}{L} \mu_{\text{inv}} C_{ox} (V_g - V_{th})^2 / 2 = 0.28\ \text{mA}/\mu\text{m}$, versus DD $0.25\ \text{mA}/\mu\text{m}$ — agreement to $\sim 14\%$. (Rev. 2’s apparent $3.5\times$ discrepancy was an artifact of assuming $V_{th} = 0.5\ \text{V}$; the extracted value removes it, leaving only the expected small velocity-saturation reduction.)

5 Status

Phase	Component	Status
P1	Derivation (Rev. 3, twice-audited)	documented
P2–P4	Device, Poisson, DD, bands	sanity-checked
P5	First-order SHE core ($(\mathbf{r}, H)$ form)	in progress
P6	Scattering: bulk prototype validated; 2-D production	prototype / pending
P7–P9	Impact ionization, self-consistent loop, Figs. 2–8	pending

Table 2: Foundation checks at $T = 300$ K, Boltzmann statistics.

Quantity	Expected	Computed	Note
<i>(A) Analytic benchmarks</i>			
N_c nonparabolic (converged grid)	2.86×10^{19}	2.857×10^{19}	auditor-independent; 0.1%
N_c parabolic: code vs. closed form	2.726×10^{19}	2.723×10^{19}	prefactor 0.09%
$v(0.1 \text{ eV}), v(1.0 \text{ eV})$ [cm/s]	—	$3.09/6.42 \times 10^7$	auditor-reproduced
V_{bi} ($n^+/10^{16}$) [V]	0.933	0.933	$V_t \ln(N_D N_A / n_i^2)$
n^+ contact potential [V]	0.586	0.586	$V_t \ln(N_D / n_i)$
bulk T_e at zero field [K]	309.7	311	[Rev.3: B1] non-parab. Maxwellian moment; $\rightarrow 309.7$ as $\Delta\epsilon \rightarrow 0$
optical operator detailed balance	0	2×10^{-5}	Maxwellian null
Maxwellian recovered at $E=0$	exact	3×10^{-4}	max rel. dev., $\epsilon < 0.4 \text{ eV}$
<i>(B) Construction targets (built-to, not validation)</i>			
metallurgical junction x_j [μm]	0.17 (target)	0.170	retuned to this value
Newton convergence	quadratic	quadratic	to 10^{-11} , 6 steps
<i>(C) Qualitative sanity trends</i>			
depletion width, abrupt vs. graded [μm]	0.347	0.269	graded junction narrower (not an equality)
bulk T_e vs. field (0/100/300 kV/cm) [K]	monotone $\uparrow$	311/1478/3392	hot-electron heating

A Numerical and model specifications

Constants: $T = 300$ K, $V_t = 25.85$ mV, $n_i = 1.45 \times 10^{10} \text{ cm}^{-3}$, Boltzmann statistics, $\epsilon_{si} = 11.7$, $\epsilon_{ox} = 3.9$, $m^* = 0.32m_0$, $g_v = 6$, $\alpha = 0.5 \text{ eV}^{-1}$, $v_s = 9.04 \times 10^3 \text{ m/s}$. **Doping (erfc profiles), lengths in μm :** donors $N_D = 10^{20}[L(x; 0.15)D(y; 0.105) + R(x; 0.65)D(y; 0.105)] + 10^{18}[L(x; 0.22)D(y; 0.06) + R(x; 0.58)D(y; 0.06)]$ where $L(x; e) = \frac{1}{2}\text{erfc}((x - e)/0.028)$, $R(x; e) = \frac{1}{2}\text{erfc}((e - x)/0.028)$, $D(y; x_j) = \frac{1}{2}\text{erfc}((y - x_j)/0.025)$; acceptors $N_A = 10^{16} + 4 \times 10^{17} \cdot \frac{1}{2}[\text{erfc}((x - 0.62)/0.028) - \text{erfc}((x - 0.18)/0.028)]D(y; 0.06)$. Gate $x \in [0.20, 0.60]$; source contact $x \leq 0.15$, drain $x \geq 0.65$ (top); body = bottom row. **Mesh:** tensor grid, surface-refined $y \propto s^{1.8}$; figures use $N_x \times N_y = 90 \times 80$ (Poisson/bands), 80×70 (DD bias), 56×48 (V_{th} sweep). ϵ -grid uniform $\Delta\epsilon = 2.5 \text{ meV}$ (bulk); global H -grid per §2.1. **Poisson BCs:** ohmic Dirichlet $\phi = V + V_t \text{asinh}(N_{net}/2n_i)$; gate oxide Robin BC: transmissibility $T_{ox} = \epsilon_{ox}\epsilon_0\Delta x_i/t_{ox}$ ($t_{ox} = 9.6 \text{ nm}$) to a metal node at $V_g - \Phi_{gate}$, $\Phi_{gate} = 0.30 \text{ V}$; Neumann elsewhere. Newton clip 0.5 V , tol 10^{-8} V . **Continuity:** SG current $J_{n,a \rightarrow b} = \frac{q\mu_n V_t}{h}[B(\Delta)n_b - B(-\Delta)n_a]$, $B(x) = x/(e^x - 1)$; ohmic $n = \frac{1}{2}(N_{net} + \sqrt{N_{net}^2 + 4n_i^2})$, $p = n_i^2/n$. **Mobility:** Caughey–Thomas (electrons $\mu_{\min, \max}/N_{ref}/a = 68.5/1414/9.2 \times 10^{16}/0.711$; holes $44.9/470.5/2.23 \times 10^{17}/0.719$), field saturation $v_{sat}^n = 1.07 \times 10^7 \text{ cm/s}$, $\beta_n=2$; $v_{sat}^p = 8.37 \times 10^6 \text{ cm/s}$, $\beta_p=1$. **SRH:** $\tau_n = \tau_p = 10^{-7} \text{ s}$, semi-implicit. **Gummel** tol 10^{-5} V . **Optical operator:** $ZQ_{op} = c_{op}Z[(N+1)Z_+f_+ + NZ_-f_- - ((N+1)Z_- + NZ_+)f]$, $c_{op} = 4.69 \times 10^{-34} \text{ J m}^3/\text{s}$ (from $\tau_{op}^{ref} = 10^{-13} \text{ s}$), $\tau_1 = 10^{-14} \text{ s}/(\sqrt{\gamma}\gamma')$. **Contact current:** drain electron current = $\sum$ SG fluxes into drain nodes, per μm ($\times 10^{-4}$ from A/cm). **SHE contacts (Part II):** $f_0 = \text{normalized Maxwellian } Z^{-1}n_{contact} e^{-\epsilon/k_B T} / \int Z e^{-\epsilon/k_B T} d\epsilon$.

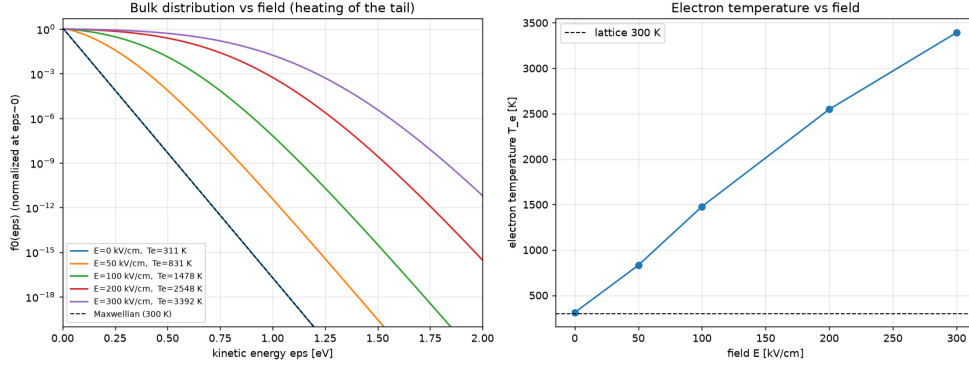


Figure 2: [Rev.3: B1] Bulk validation after adding the acoustic energy operator: zero field now reproduces the Maxwellian (moment $T_e = 311$ K $\rightarrow$ 309.7 K on refinement); the tail heats monotonically with field. This is a *bulk-prototype* validation; the 2-D production scattering implementation is pending (P6).

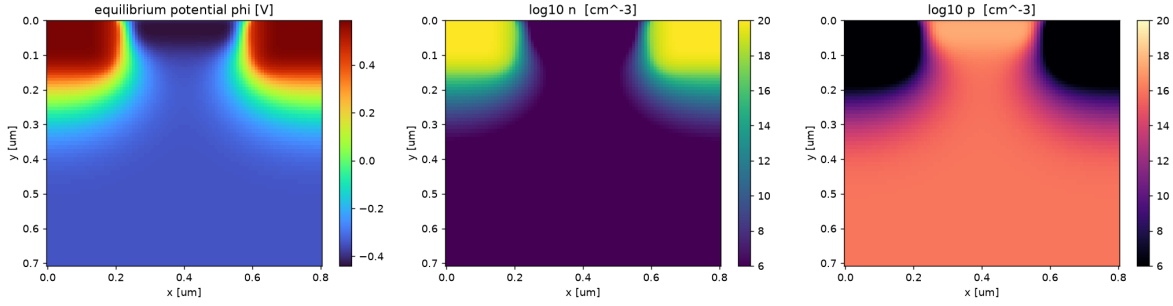


Figure 3: Equilibrium Poisson: potential, $\log_{10} n$, $\log_{10} p$.

B Code and parameter provenance

Git commit 4660551; Python 3.13.14, NumPy 2.5.1, SciPy 1.18.0. Modules: `constants`, `device`, `poisson`, `dd`, `bands`, `scattering`, `she_bulk`, `diagnostics`, `vth_extract`. Each figure and benchmark is regenerated by running the corresponding module.

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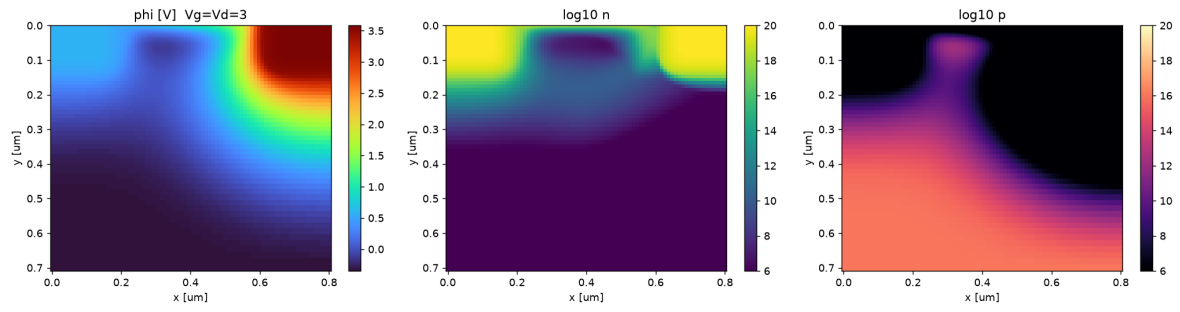


Figure 4: Self-consistent DD at $V_{gs} = V_{ds} = 3$ V: inversion layer and drain-side high-field / pinch-off region.

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